Smart Clinic Database System

Mid-Project Progress Report

IT244 - Database Design and Implementation

Student Name	Student ID	Primary Contribution
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Cyrine Abdullah Alghamdi	240015574	Task 1: Database Design and EER Documentation
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1. Progress Summary

The project has completed the core design, implementation, and SQL evidence stages. The MySQL 8.0 database now contains eleven normalized tables, fictional Saudi sample data, the required SELECT, JOIN, nested, GROUP BY, UPDATE, DELETE, VIEW, and TRIGGER operations, and transaction-based demonstrations using ROLLBACK. The complete script was executed in MySQL Workbench, table population and query outputs were verified, and screenshots were captured for the assessed operations. The public GitHub repository is active and is being organized with the database script, EER diagram, documentation, evidence folders, and a real commit history. Final report review, repository completion, and submission packaging remain in progress.

Workstream	Status	Evidence/Output
Task 1 - Design	Completed and reviewed	EER diagram, relational schema, keys, cardinalities, assumptions
Task 2 - Implementation	Completed in MySQL	Database, eleven tables, constraints, and Saudi sample records
Task 2 Evidence	Captured and verified	Workbench code, populated table outputs, and execution status
Task 3 - SQL Operations	Completed in MySQL	SELECT, JOIN, nested, GROUP BY, UPDATE, DELETE, VIEW, and TRIGGER
Task 3 Evidence	Captured and verified	Actual Workbench results, rollback tests, and stock-deduction evidence
Repository and Documentation	In progress	README, SQL script, ERD, evidence folders, report files, and commits

2. Completed Work

1. Defined eleven normalized tables: person, patient, employee, doctor, receptionist, appointment, treatment, medicine, prescription, prescription_item, and payment.
2. Modeled PERSON as a supertype specialized into PATIENT and EMPLOYEE, with EMPLOYEE further specialized into DOCTOR and RECEPTIONIST.
3. Specified primary keys, foreign keys, uniqueness constraints, checks, controlled ENUM values, and referential actions.
4. Produced a complete EER diagram with entity attributes, subtype structures, relationships, and cardinalities.
5. Inserted at least five logically connected fictional Saudi records into every required table.
6. Executed the required SELECT, JOIN, nested, GROUP BY, UPDATE, DELETE, VIEW, and TRIGGER operations in MySQL Workbench.
7. Created the reporting view vw_appointment_summary and the stock-control trigger trg_prescription_item_before_insert.
8. Used transactions and ROLLBACK to demonstrate reversible appointment deletion and medicine-stock changes safely.
9. Captured MySQL Workbench screenshots and established a public GitHub repository with organized files and meaningful commits.

3. Key Design Decisions

Decision	Reason
Shared PERSON supertype	Avoids repeating identity and contact attributes across patients and employees.
EMPLOYEE subtype hierarchy	Separates common employment data from doctor-specific and receptionist-specific attributes.
Associative prescription_item table	Stores dosage, frequency, duration, and quantity while resolving the prescription-medicine relationship.
Multiple payments per appointment	Supports partial payments, insurance portions, and different payment methods.
Unique doctor schedule and controlled statuses	Reduces scheduling conflicts and preserves consistent appointment and payment states.
Single MySQL script with stock trigger	Keeps implementation reproducible and prevents prescription quantities from exceeding available stock.

4. Challenges and Responses

Implementation Challenge	Response
Representing specialization relationally	Used PERSON and EMPLOYEE as supertypes and made subtype primary keys foreign keys to their parent tables.
Maintaining referential insertion order	Created and populated parent entities before dependent appointment, treatment, prescription, and payment records.
Adapting the sample data to Saudi context	Replaced earlier data with fictional Saudi names, +966 phone formats, Gmail addresses, and Saudi cities.
Testing UPDATE and DELETE safely	Wrapped demonstrations in transactions and used ROLLBACK so the final dataset remained unchanged.
Validating the stock trigger and evidence	Captured before-and-after stock values, confirmed the trigger deduction, and rolled back the test afterward.

5. Team Contributions

Member	Current Contribution	Status
JOMANAH SAEED OTAIF	Project coordination, repository management, README updates, file organization, integration review	In progress
Cyrine Abdullah Alghamdi	Entity analysis, keys, relationships, cardinalities, EER structure, assumptions, and normalization review	Completed and reviewed
Leen Sultan Al Shmmari	MySQL schema review, required SQL operations, view, trigger, transactions, and execution verification	Completed and reviewed
Bedour Hamad Alrasheedi	Workbench screenshots, testing evidence, report formatting, quality checks, and submission checklist	In progress
All members	Cross-review, final report approval, contribution verification, and final submission preparation	Ongoing

6. Next-Stage Plan

1. We will complete the remaining GitHub folders and upload the report, contribution record, and screenshot evidence using meaningful real commits.
2. We will review the final EER diagram, relational schema, constraints, sample data, and all assessed SQL outputs as a team.
3. We will verify that every screenshot is readable, correctly named, and matched to the relevant Task 2 or Task 3 requirement.
4. We will update the final report with the verified GitHub repository link, Google Docs collaboration evidence, and completed team-contribution table.
5. We will complete the project reflection after confirming the final technical results, challenges, and lessons learned.
6. We will inspect the final DOCX and PDF, confirm all personal and course details, and prepare the compressed submission package.